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!pip install --quiet opencv-python-headless
!pip install --quiet matplotlib
import cv2, numpy as np, matplotlib.pyplot as plt
img = cv2.imread('rgb.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)); plt.title('RGB Image')
plt.figure(); plt.imshow(gray, cmap='gray'); plt.title('Grayscale')
plt.figure(); plt.hist(gray.ravel(), 256); plt.title('Histogram')
for i in range(8):
    plt.figure(); plt.imshow(((gray >> i) & 1)*255, cmap='gray')
    plt.title(f'Bit Plane {i}')
```